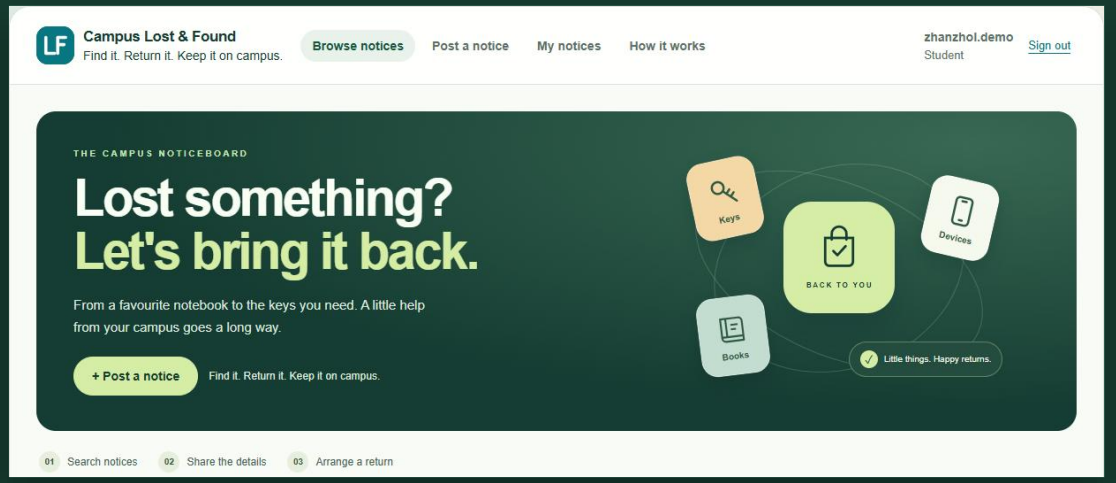
ICT FINAL PROJECT · SE-2623 · WEEK 10

# Lost it on campus? Find it in one place.

Campus Lost & Found — the finished, tested system, and what three of us learned building it.




Real screenshot of the final build · synthetic demo data · runs locally

Mengdigerey Zhanzhol
Galitskiy Daniil
Bisengaliev Daulet

# Lost items wait for days because notices are scattered



# 4 of 10

students in our survey said getting an item back took more than three days

**9 of 10** said a shared, searchable board would be useful

Stage 1 survey, October 2026 · n = 10 · convenience sample of fellow students — a signal that the problem exists, not a measurement of its size.

| TODAY | | WITH CAMPUS LOST & FOUND |
|---|---|---|
| **Group chats**<br>posts scroll away | → | **One board**<br>search by title and category |
| **Paper notes**<br>only seen in one corridor | → | **University accounts**<br>everyone knows who posted |
| **Guard desk**<br>no one knows it is there | → | **Email the author**<br>one click from the notice |
| **No status**<br>is it already returned? | → | **Returned badge**<br>closed notices stay honest |

# All 11 Must requirements are done



FUNCTIONAL (F) · F3 is the only Should

| F1 | F2 | F3 | F4 | F5 | F6 | F7 | F8 |
|---|---|---|---|---|---|---|---|
| done | done | done | done | done | done | done | done |
| University sign-in | Publish | One photo | Browse + details | Search + filter | Email author | Edit + returned | Admin removes |

NON-FUNCTIONAL (N)

| N1 | N2 | N3 | N4 |
|---|---|---|---|
| done | done | done | done |
| Search ≤ 2 s | Identity + rights | 360 px + keyboard | 30-min demo |

## University sign-in now works

Students sign in with their Astana IT University Microsoft account (OpenID Connect + PKCE). Verified with a real account on 8 Nov 2026.

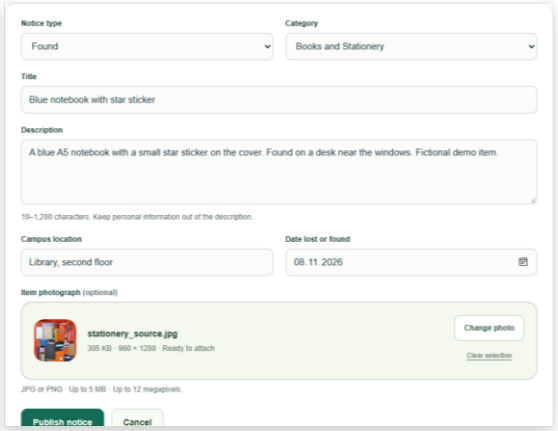
Only @astanait.edu.kz accounts from the university directory get in; in this mode nobody can self-register with an unverified address. Demo accounts stay for offline demos.

**Evidence for every row**
Final/Requirements/Requirements_Status.md lists the test, screenshot or measurement behind each status. All were re-run on 8 Nov 2026 on the final code.

# Live demo · a student goes from “I found it” to “Returned”

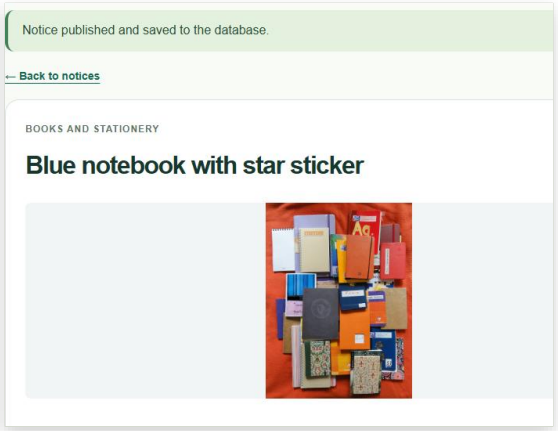
Zhanzhol · 1:15

LIVE DEMO



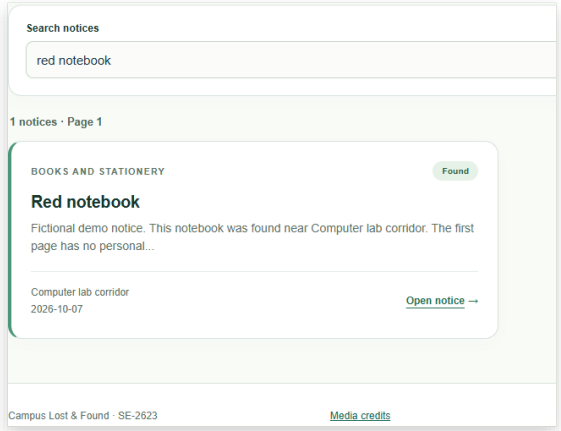

## 1 Publish

Type, category, title, place, date, one photo



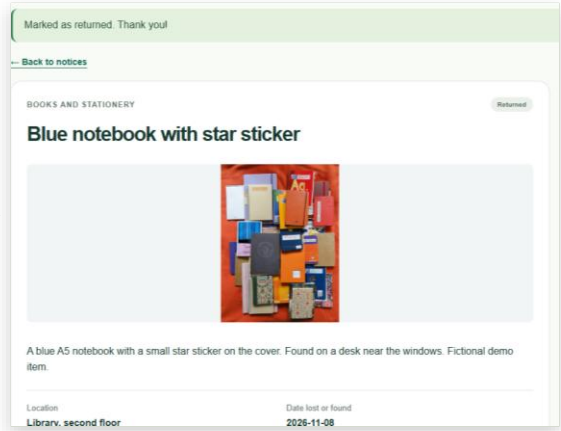

## 2 Saved

Stored in SQLite; photo re-encoded and private




## 3 Find

Search “red notebook” + category → 1 result




## 4 Close

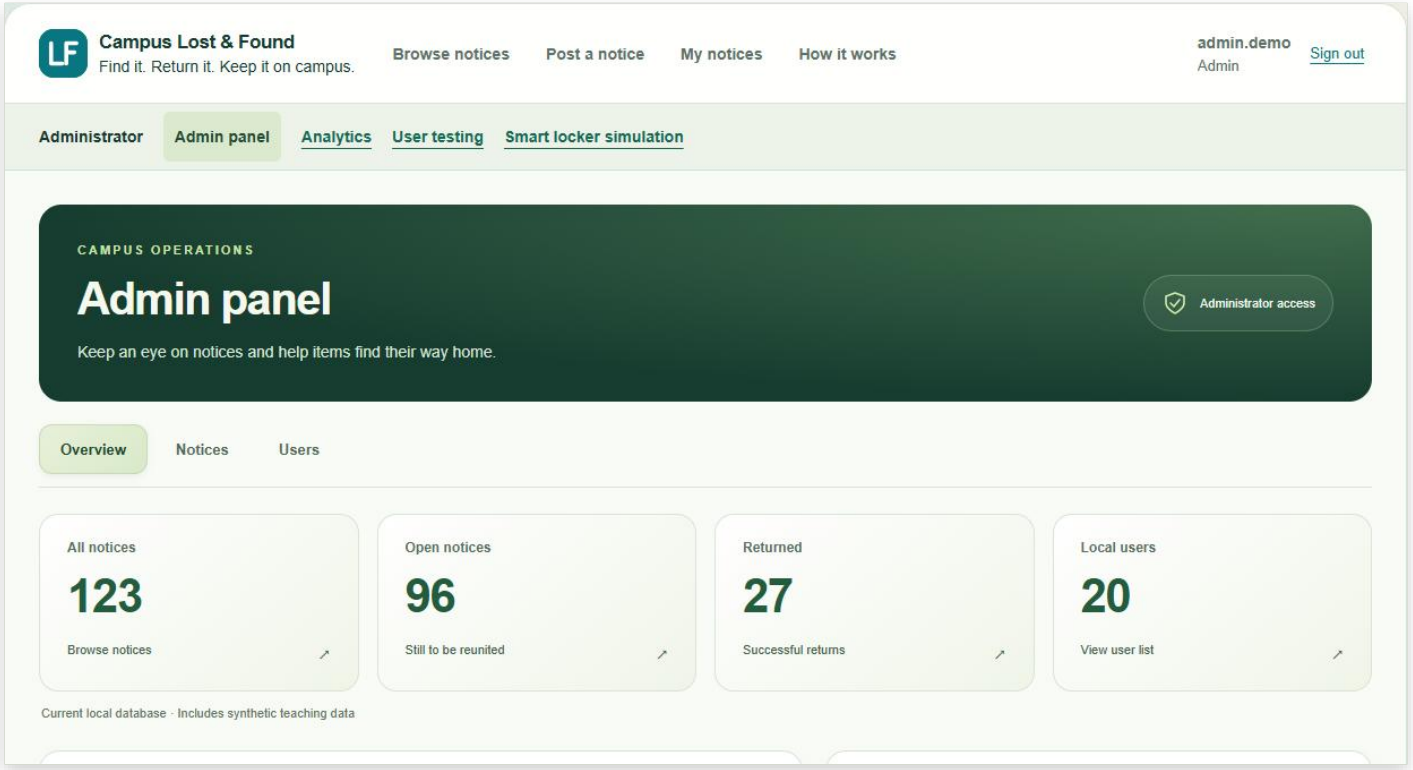
Email the author, then mark it Returned

All four frames are real captures of the final build (8 Nov 2026). If the live demo fails, talk through these frames.

# Live demo · the admin keeps the board clean

Daniil · 1:10

LIVE DEMO




1 **Live counts from SQLite**
Notices, open, returned and users are counted on every visit — nothing is typed in by hand.

2 **Remove with a confirmation step**
Opening the page deletes nothing; only "Confirm removal" does. A student gets 403.

3 **Analytics on the same data**
Three SQL questions with CSV export, clearly labelled "synthetic teaching data".




Admin pages return 403 to all three student accounts and 200 to the administrator (checked 8 Nov 2026).

# The final code passed every automated and security check

Daniil · 0:50

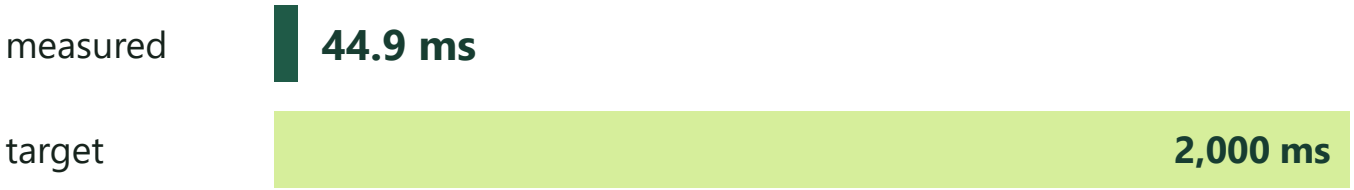
**96 / 96**

automated tests passed on the final code — logins, CSRF, ownership, photos, analytics, backup, locker

Ran 96 tests in 42.2 s — OK · 8 Nov 2026

**These are software checks. They are never counted as human testers — that is the next slide.**

### N1 · Search response, 10 users at once (95th percentile)




Same scale. 100 searches from 10 signed-in sessions over local HTTP; page images and drawing time not included.

**5 / 5**

invalid-access attempts blocked (SC3): wrong domain, wrong password, anonymous, foreign edit, student removal

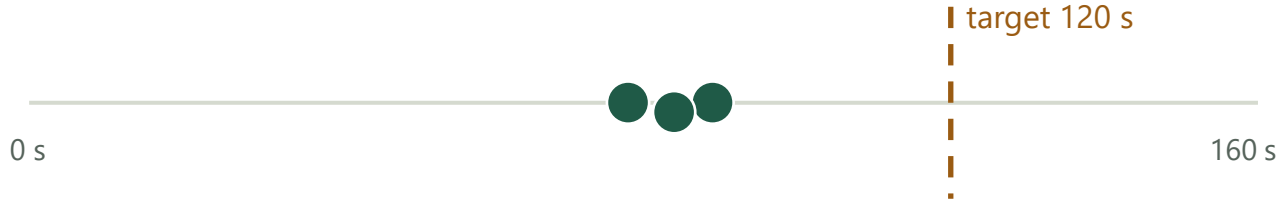
**20 / 20**

availability checks passed, spread over 30 minutes (N4), plus offline → retry in the browser

# 3 people outside the team re-tested our user criteria

Daniil · 0:45

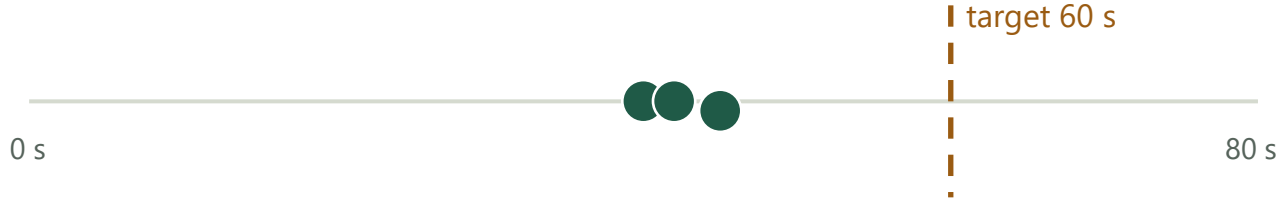
**SC1 · Publish**




3/3

**met (≥ 80 %)**

**SC2 · Find**




3/3

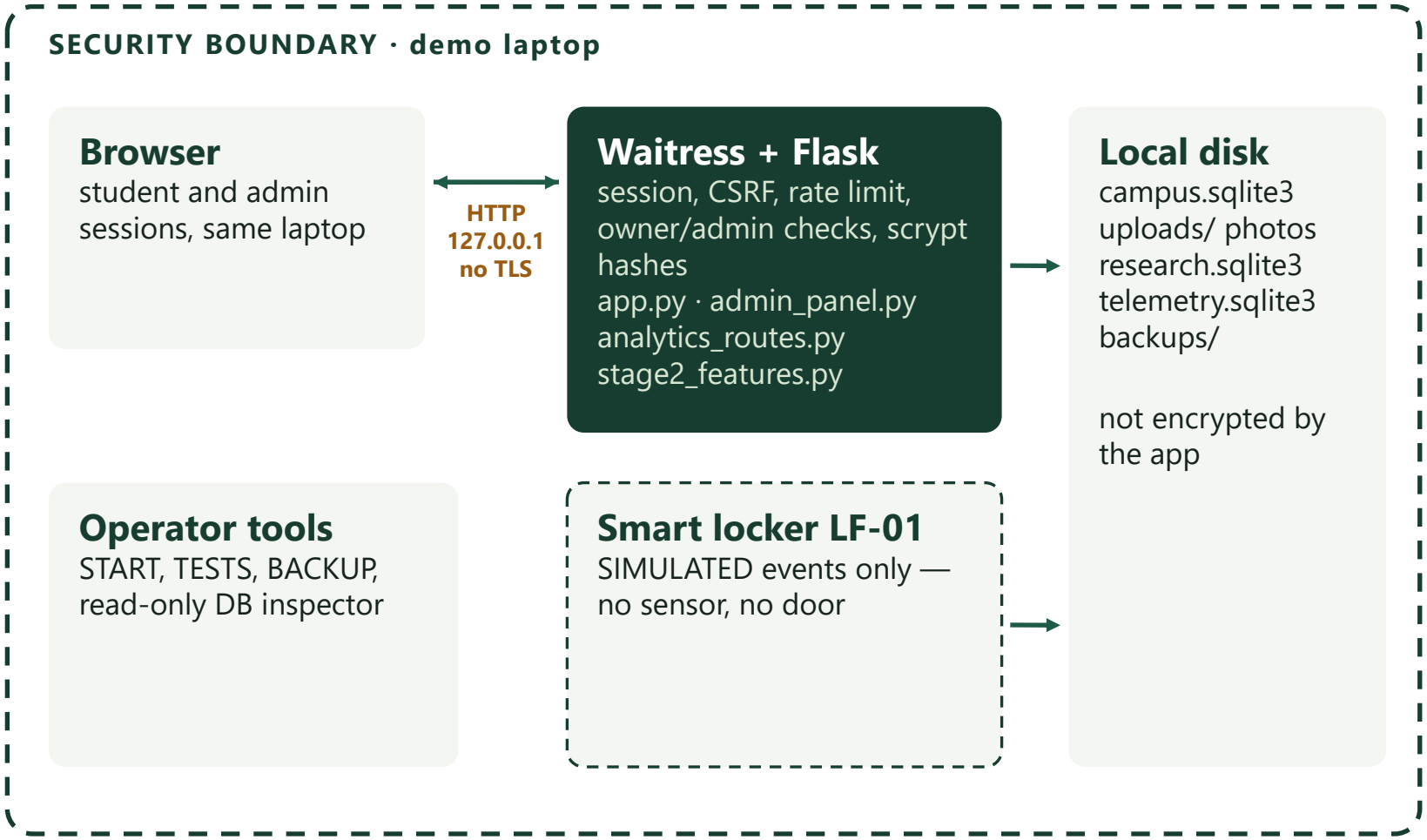
**met (≥ 80 %)**

| Criterion | Stage 2 result | Final result | Met? |
|---|---|---|---|
| SC1 · post within 2 min (≥ 4/5) | Report: Pending · team-reported 5/5, no raw record | 3/3 | ✓ Yes |
| SC2 · find within 1 min (≥ 4/5) | Report: Pending · team-reported 5/5, no raw record | 3/3 | ✓ Yes |
| SC3 · invalid access blocked (5/5) | 5/5 | 5/5 (8 Nov) | ✓ Yes |

Each dot is one person's first attempt (green = correct, unassisted, within target). 3 people outside the team, 2026-11-06 and 2026-11-07; Windows 11 / Chrome, Windows 11 / Microsoft Edge, Android / Chrome. Small sample: one person = 33 percentage points.

# One laptop; the security boundary is its Windows account






## OUTSIDE THE BOUNDARY

**Other devices**
cannot connect — server binds 127.0.0.1

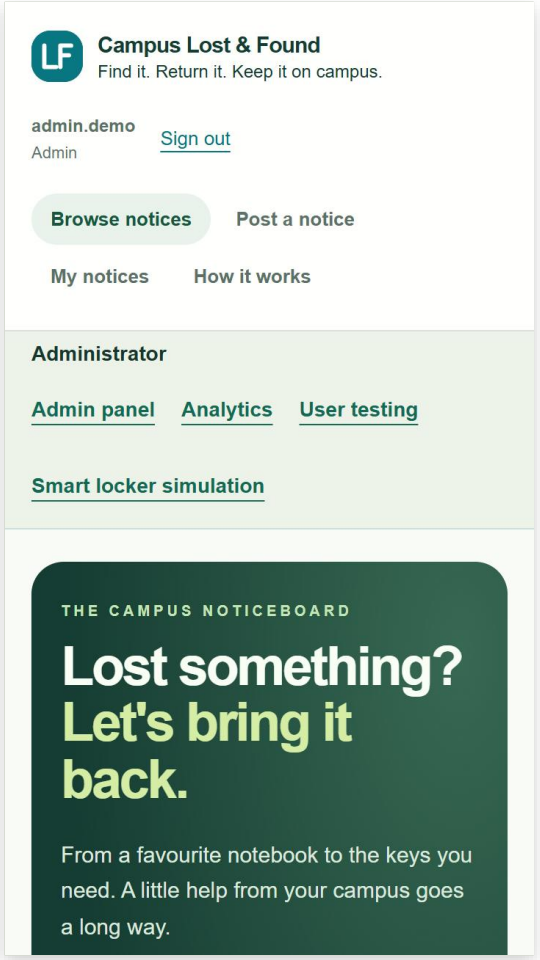
**Email app**
"Contact author" opens a draft; site sends nothing

**Microsoft Entra ID**
university sign-in (OIDC + PKCE), verified

**GitHub**
public source with synthetic data only

# Works on a 360 px phone, by keyboard and after a drop



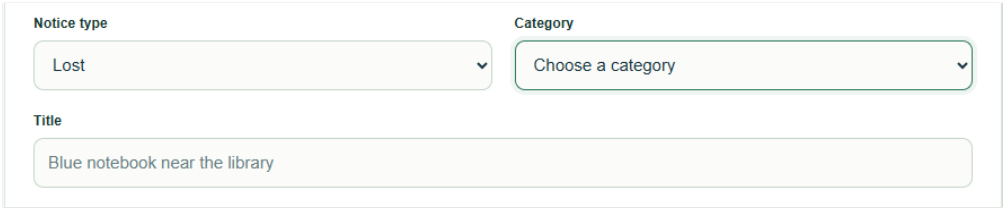



## 12 / 12 pages

no sideways scrolling at 360 × 760 px — student, admin, sign-in and locker pages

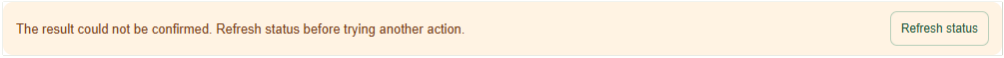
## Keyboard

Tab follows the screen order to “Publish notice”; every field has a label; focus is visible




## Offline → retry

network cut: a clear message, no double submit; reconnect: “Status refreshed”

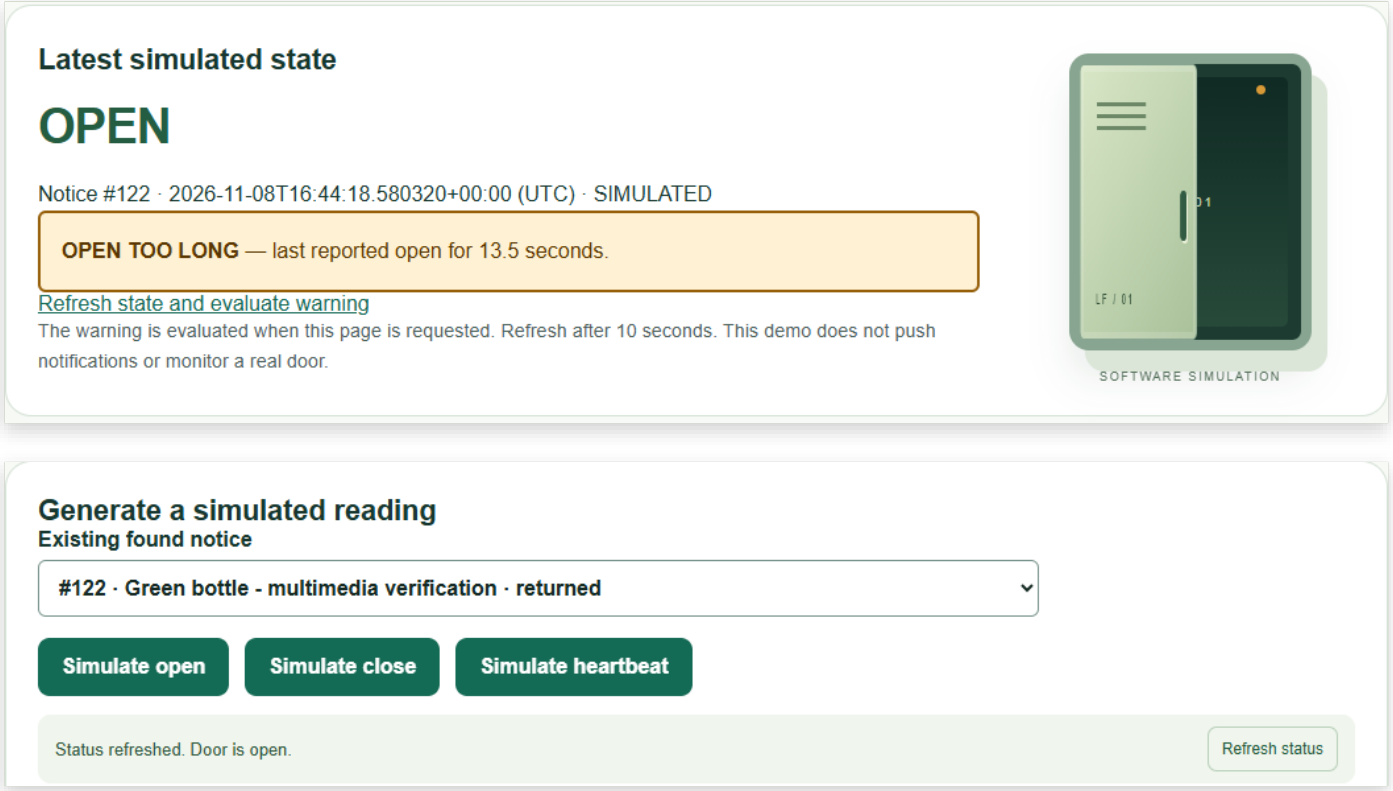


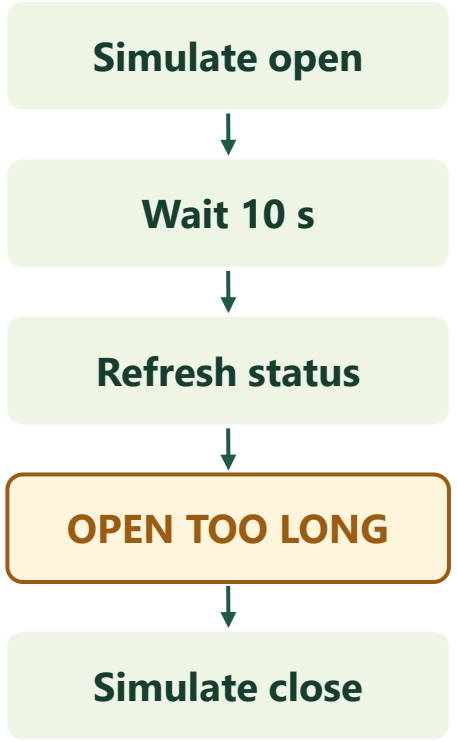

Automated checks in Microsoft Edge on 8 Nov 2026 — not an accessibility audit with real assistive-technology users.

# Live demo · a simulated locker warns when it is left open

Daulet · 0:55

LIVE DEMO




**SOFTWARE SIMULATION**

No sensor, no door, no network device. Every reading is stored as SIMULATED and a notice is never closed automatically — a person checks the locker.

Buttons update the page in place (fetch with CSRF token, no reload). Physical version proposed: reed switch + Wi-Fi controller, HTTPS, device keys.

# What we would build next, and what we learned



| LIMIT NOW | NEXT STEP |
| --- | --- |
| **Sign-in only on localhost** | → HTTPS redirect, secret in a vault, store the immutable oid claim |
| **One laptop, local HTTP** | → Host in a Kazakhstan data centre with HTTPS — personal data must stay in Kazakhstan |
| **SQLite, one writer** | → PostgreSQL + object storage before 10,000 users |
| **Free-text locations** | → A fixed list of campus places for cleaner analytics |

## Lessons

- Evidence beats claims — the requirement table separated "passes a test" from "a person used it".
- Cutting chat and matching kept the core flow stable; university sign-in came last, once the core worked.
- One upload commit hid our individual work — next time, one commit per feature.
- Start testing with real users in week 7, not week 10.

# Thank you

Questions — about any part of the system — are welcome.

**Repository**
github.com/zondbebox/SE-2623_Endterm · branch main · commit a7e5c74

**Run it**
Campus_Lost_Found\START.bat → http://127.0.0.1:5055

**Demo accounts (synthetic)**
student zhanzhol.demo@astanait.edu.kz · admin admin.demo@astanait.edu.kz · shared demo password in the README

Mengdigerey Zhanzhol
Galitskiy Daniil
Bisengaliev Daulet

# Backup · if the live demo fails, we keep going



| | |
|---|---|
| **Server stopped** | Close the window, double-click START.bat again, sign in (30 s). |
| **Port 5055 busy** | Stop the old window with Ctrl+C, or run python\python.exe app.py --port 5056. |
| **Laptop or browser fails** | Narrate slides 4, 5 and 10 — every frame is a real capture of the final build. |
| **Need motion** | Local MP4 Video/SE-2623_Campus_Lost_Found_A4.mp4 (2:14) or youtu.be/76M3hSqR-7w — older interface, student flow only; say so. |

# Backup · where personal data lives, and why the law cares

## Where it is now

Emails, password hashes and notices: data/campus.sqlite3. Photos: data/uploads/. Backups: backups/. All on the demo laptop's disk, in Kazakhstan, not encrypted by the app. GitHub holds synthetic data only.

## What the law says

Law RK No. 94-V "On Personal Data and Their Protection": Art. 12(2) storage in a database located in Kazakhstan; Arts. 7–8 consent; Art. 22 protection; Art. 25(2)(8) breach notice within one working day (Russian text, as amended 14 Sep 2026).

## If we went public

Kazakhstan data centre · HTTPS · consent text · retention period · incident procedure. We do not claim legal compliance.

# Backup · with 10,000 users, access fails first, then writes



**1 · Reachability**
Loopback only: nobody else can open the site

→

**2 · Writes**
SQLite = one writer; Waitress = 4 threads; uploads up to 6 MB queue

→

**3 · Storage**
Photos on one disk; in-memory rate limit resets on restart

→

**4 · Identity**
Local accounts cannot prove 10,000 real students

**Plan:** host in Kazakhstan with HTTPS → PostgreSQL → object storage for photos → several app instances behind a load balancer with a shared rate limiter → HTTPS redirect for Entra sign-in → load-test before launch. Our 45 ms p95 is for 10 users on a laptop and does not predict 10,000.

# Backup · sources for the facts in this talk

- Stage 1 report (9 Oct 2026) — survey n = 10, requirements F1–F8, N1–N4, SC1–SC3
- Stage 2 report (1 Nov 2026) — Stage 2 statuses; SC1/SC2 recorded as Pending
- Final/Evidence (8 Nov 2026) — test_suite_final.log, search_performance_final.json, sc3_final.json, stability_30min_final.json, browser_checks.json, account_checks.json, screenshots
- Final/Testing — Final_Test_Protocol.md, final_test_results.csv, stage2_team_reported.csv
- Law RK No. 94-V, adilet.zan.kz (checked 8 Nov 2026); Microsoft Learn — Entra app registration and OpenID Connect; SQLite — Appropriate uses; OWASP cheat sheets
- Code: github.com/zondbebox/SE-2623_Endterm, commit a7e5c74 (only commit, uploaded from the zondbebox account)